

Has the evidence for moral licensing been inflated by publication bias?

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Abstract

Moral licensing describes the phenomenon that displaying moral behavior can lead to subsequent immoral behavior. To explain this effect, it has been suggested that an initial moral act affirms the moral self-image and hence licenses subsequent immoral acts. Previous meta-analyses on moral licensing indicate significant overall effects of $d > .30$. However, several large replication studies have either not found the effect or have reported a substantially smaller effect size. The present article investigated whether this can be attributed to publication bias. Datasets from two previous meta-analyses on moral licensing were compared and discrepancies in effect size coding, sample size coding and study inclusion were corrected. The larger and more recent dataset was used for the present analyses. Using PET-PEESE and a three-parameter-selection-model (3-PSM), we found strong evidence for publication bias. Furthermore, the adjusted effect sizes were reduced to $d = -.10$ and $d = .10$, respectively. We conclude that both the evidence for and the size of moral licensing effects have been inflated by publication bias. Future studies on the effect should be adequately powered and ideally pre-registered.

Has the evidence for moral licensing been inflated by publication bias?

Moral licensing theory postulates that initially displaying a moral action increases the probability of subsequent behavior that is immoral, unethical or otherwise problematic (Merritt, Effron, & Monin, 2010). This phenomenon has been investigated in several studies across various life domains. For instance, previous gender-egalitarian acts were demonstrated to result in subsequent gender-discriminatory behavior (Monin & Miller, 2001). Similarly, participants who were initially given the chance to purchase green products subsequently shared less money in an interpersonal interaction task and stole more money in a self-gratification paradigm (Mazar & Zhong, 2010). The moral licensing effect appears to be at odds with several prominent psychological findings and theories which imply a human striving for consistency (Blanken, van de Ven, Zeelenberg, & Meijers, 2014). For instance, self-perception theory (Bem, 1972) posits that people constantly observe their own behavior in order to draw conclusions about their attitudes. These inferred attitudes strongly influence subsequent behavior and are thought to establish consistency. Drawing upon a moral self-regulation framework, the inconsistency in behavior described as moral licensing was explained as an internal balancing of the moral self-image and the costs which are associated with moral deeds (Sachdeva, Iliev, & Medin, 2009). Specifically, by verifying the moral self-image, the prior moral action was suggested to provide a license for ensuing immoral behavior. Conversely, a moral transgression was reasoned to evoke an ensuing compensatory moral action in order to reinstate this internal balance (Sachdeva et al., 2009). Thus, the moral licensing effect could be argued to reflect the outcome of constant moral cost-benefit calculations.

Moderators of the moral licensing effect are largely unknown (Blanken, van de Ven, & Zeelenberg, 2015), but Simbrunner and Schlegelmilch (2017) recently provided evidence for two potential moderators: (1) The culture of the studied population and (2) the type of control condition (neutral vs. immoral previous behavior). Specifically, they argued that the cultural background of participants influences their moral self-concept and should thus moderate the moral licensing effect (Simbrunner & Schlegelmilch, 2017). Additionally, they showed that the effect was larger when a previous moral behavior was contrasted with a previous immoral behavior instead of a neutral control condition.

Replication issues

Two recent meta-analyses have been conducted on the moral licensing effect. Blanken et al. (2015) found an effect size of $d = .31$ over 91 studies and more recently, Simbrunner and Schlegelmilch (2017) reported an overall effect size of $d = .32$ over 106 studies. Despite

these seemingly encouraging findings, there have been concerns about the replicability of moral licensing effects. Specifically, two high-powered sets of studies ($n = 801$ and $n = 1274$) found no evidence for a moral licensing effect (Blanken et al., 2014; Urban, Bahník, & Kohlova, 2017). An even larger replication attempt was conducted as part of a many-labs replication project. The study found the expected licensing effect with $n = 3134$, but the effect size was less than half of the estimates derived from recent meta-analyses ($d = .15$; Ebersole et al., 2016).

How can these replication issues and reduced effect sizes in large studies be reconciled with the results of the meta-analyses? This question touches upon the larger issue of a “crisis of reproducibility” in psychology (Pashler & Wagenmakers, 2012). The replicability of psychological studies has recently been empirically estimated to be only around 40%, although this estimate varies across subfields of psychology (Open Science Collaboration, 2015). In addition to this, questions were raised about the replicability of several well-established effects that were studied in hundreds of experiments. A well-known example of this is the ego depletion literature (Baumeister, Bratslavsky, Muraven, & Tice, 1998), where a meta-analysis over almost 200 studies reported an effect size of $d = .62$ (Hagger, Wood, Stiff, & Chatzisarantis, 2010). However, a re-analysis indicated that the effect is not as robust as previously thought (Carter & McCullough, 2014; see below) and two recent highly powered replication projects found no evidence for an effect (Hagger et al., 2016; Etherton et al., 2018).

At least two core reasons have been identified for the surprising lack of reproducibility in some areas of psychology: (1) publication bias - the preferential publication of significant results and (2) questionable research practices (QRPs) that inflate the type-1-error rate. Publication bias (Rothstein, Sutton, & Borenstein, 2006) has been a well-known problem in psychology for several decades (e.g., Greenwald, 1975) and the replication crisis has raised awareness for the issue (e.g., Ferguson & Heene, 2012; Francis, 2012; Franco, Malhotra, & Simonovits, 2014; Ioannidis, Munafo, Fusar-Poli, Nosek, & David, 2014). It represents a major threat to conventional meta-analyses given that publication bias which is unaccounted for can lead to type-1-error rates close to 100% and substantial effect size estimates in the absence of an effect (e.g., Carter, Schönbrodt, Gervais, & Hilgard 2018). In addition, researcher degrees of freedom allow for QRPs that artificially decrease p-values, with the explicit or implicit goal to obtain $p < .05$. Examples of such QRPs include optional stopping, dropping experimental conditions, changing the dependent variable and selective exclusion of outliers (Simmons, Nelson, & Simonsohn, 2011).

Testing and correcting for publication bias

In order to test whether publication bias can explain recent replication issues of moral licensing, we attempt to both detect and correct for publication bias in the moral licensing literature. Blanken et al. (2015) have already provided evidence for publication bias by regressing effect sizes on their standard errors. They found larger standard errors were associated with larger effect sizes, which suggests the presence of publication bias (Egger, Smith, Schneider, & Minder, 1997; see below). However, they restricted this analysis to published studies only instead of considering the entire dataset. This approach might miss publication bias that persisted even after the inclusion of some unpublished studies. In addition, they have graphically used the trim and fill method (Duval & Tweedie, 2000) to correct for publication bias, which has been criticized (Moreno et al., 2009). Finally, they showed that the mean effect size of published studies is larger than that of unpublished studies, again suggesting the presence of publication bias (Blanken et al., 2015). Contrary to this, Simbrunner and Schlegelmilch (2017) showed that the effect of publication status disappeared in a moderator-analysis which included other predictors of moral licensing. Using Rosenthal's (1979) fail-safe N, they demonstrate that 4531 nonsignificant unpublished studies would be required to reduce the overall result to nonsignificance. Simbrunner and Schlegelmilch (2017) conclude that their result is robust to publication bias.

We seek to expand on these initial statistical tests for publication bias in the moral licensing literature by testing and correcting for publication bias in the entire dataset including unpublished studies. Since not all unpublished studies can be obtained in meta-analyses, datasets including unpublished work are still likely to be influenced by publication bias. We implemented methods to correct for publication bias which have shown to perform best in simulation studies. The previously employed methods trim and fill and fail-safe N have both been shown to possess less than optimal properties. Specifically, trim and fill substantially overestimates the true effect size in the case of publication bias (Moreno et al., 2009; Carter et al., 2018). Fail-safe N, on the other hand, suffers from at least two issues: (1) QRPs that increase the type-1-error rate artificially increase the estimated fail-safe N (Simonsohn, Nelson, & Simmons, 2014) and (2) fail-safe N is merely concerned with statistical significance and not effect sizes. Meta-analyses, however, are usually concerned with effect size estimation and not just the test of (nil-)null hypotheses. The use of fail-safe N has thus been discouraged in favor of other available methods (Becker, 2005). We will focus on two such methods: PET-PEESE and the three-parameter selection model (3-PSM).

PET-PEESE. PET-PEESE (Stanley & Doucouliagos, 2014) can be used both as a test for publication bias and as a way to correct for publication bias. It models the relationship between effect sizes and their standard errors using a weighted linear regression. A significant slope indicates the existence of publication bias given that in most cases no such relationship should exist without publication bias. The selective omission of small studies with small effect sizes, however, leads to a positive relationship between effect size and standard error. The intercept of the weighted regression provides an estimate of the corrected mean effect size¹. PET-PEESE has been famously applied to the ego-depletion (Baumeister et al., 1998) literature. Whereas a meta-analysis on ego-depletion identified a mean effect size of $d = .62$ (Hagger et al., 2010), PET-PEESE suggested an effect that is not different from zero (Carter & McCullough, 2014). In line with the results from PET-PEESE, subsequent highly-powered replications of ego depletion indeed did not find an effect (Hagger et al., 2016; Etherton et al., 2018). PET-PEESE outperforms traditional methods to correct for publication bias such as trim and fill (see simulations from Carter et al., 2018), although it can have both upward and downward biases in certain situations. For instance, it can overestimate effect sizes of zero in the case of extreme heterogeneity together with publication bias. This is worsened when only few studies are included in the meta-analysis (Carter et al., 2018; Stanley, 2017). Nevertheless, when heterogeneity and publication bias are present, it outperforms other frequently used methods including trim and fill and p-curve in terms of both effect size estimation and type-1-error control (see simulations from Carter et al., 2018).

A potential criticism of PET-PEESE is that relationships between effect size and standard error can stem from benign causes such as power calculations. If researchers correctly anticipated their effect sizes and adjust their sample size to it, PET-PEESE would correct the effect size downwards in the absence of publication bias. However, moderators of the moral licensing effect are largely unknown (Blanken et al., 2015). This reduces the probability that researchers accurately anticipate the effect size of their study (compared to other moral licensing studies) and use this estimate in power-analyses. Therefore, this caveat is unlikely to apply to the present literature. Moreover, several large studies represent exact or slightly modified replications of initial smaller studies, leading to substantially reduced effect sizes in the replications (Blanken et al., 2014; Ebersole et al., 2016; Urban et al., 2017). This

¹ If the intercept is significant, the model is re-estimated with the variance instead of the standard error as the predictor given that this model is less biased in the presence of a non-zero effect. This was not the case in the present analyses.

suggests that a relationship between the effect size and its standard error would likely be attributable to publication bias.

3-PSM. Selection models were initially proposed by Hedges (1984) and extended by Iyengar and Greenhouse (1988) as well as Vevea and Hedges (1995). The model from Iyengar and Greenhouse (1988) simultaneously estimates (1) a mean effect size, (2) between-study heterogeneity and (3) a probability that nonsignificant studies are published. It was extended by Vevea and Hedges (1995) to incorporate moderator analyses into the selection model. The 3-PSM performed better than any other method in the simulations from Carter et al. (2018). Its mean effect size and type-1-error rate were nominal or close to nominal in most cases. However, it should be noted, that in the simulations the data-generating mechanism exactly matched the model of the 3-PSM. In cases when its assumptions, such as the normal distribution of heterogeneous effect sizes, are violated, the performance might be worse (see Hedges & Vevea, 1996). Hence, we decided to implement both the 3-PSM and PET-PEESE and compare their results.

In sum, previous work has indicated the presence of publication bias in the moral licensing literature. However, this was in part restricted to a subset of the data and in part limited by the use of suboptimal methods. We want to add to the literature by analyzing the most recent Simbrunner and Schlegelmilch (2017) dataset with state of the art methods to detect and correct for publication bias. To this end, we (1) compare the Blanken et al. (2015) and the Simbrunner and Schlegelmilch (2017) datasets to confirm that the same experiments were coded identically and to correct errors if necessary, (2) replicate the Simbrunner and Schlegelmilch (2017) results and (3) test and correct for publication bias using PET-PEESE and the 3-PSM.

Comparison between the two meta-analyses

A comparison between the two meta-analyses indicated that 93% of the effect sizes ($k = 85$) associated with the same studies were identical. We identified 6 effect sizes that were coded differently in the two datasets. For discrepancies and their resolution see table 1. The largest discrepancy emerged in the case of a study (Mazar & Zhong, 2010) where an effect size of $d = .52$ was coded as $d = 3.2$ and $d = 3.5$ by Simbrunner and Schlegelmilch (2017). This error could be attributed to the fact that the authors of this study erroneously reported SE as SD (see Mazar & Zhong, 2010). The smaller effect size is consistent with the reported t -values. Regarding the sample sizes, 93% were coded to be identical. For 95% of studies, the difference in coded sample size was smaller than 2. Six studies for which the sample sizes differed emerged. For discrepancies and their resolutions see table 1.

Table 1

Discrepancies in effect size / sample size coding and study inclusion between the meta-analyses of Blanken et al. (2015) and Simbrunner and Schlegelmilch (2017) and their resolutions.

Discrepancy	Study	Action taken
ES coding & Inclusion	Blanken et al. (2014), study 3: Coded as two studies by Simbrunner and Schlegelmilch (2017). Two dependent variables were reported. An average of the two ES is similar to the ES reported in Blanken et al. (2015). This average was included.	Given that these ES are not independent, we decided to average them and code them as one. Sign reversed (see below).
ES coding	Blanken et al. (2014), study 1-3: ES were reported to be in the opposite direction of that reported by Blanken et al. (2015) and Simbrunner and Schlegelmilch (2017).	The direction of the ES was reversed in both datasets, consistent with the original Blanken et al. (2014) report.
ES coding & Inclusion	Effron, Monin, & Miller (2012), study 1: Simbrunner and Schlegelmilch (2017) included study twice with different ES.	Instead of including both dependent ES separately, we averaged them, after which the ES was identical to that of Blanken et al. (2015).
ES coding	Jordan, Mullen, & Murnighan (2011), study 2: $d = .42$ coded as $d = 1.0$ by Simbrunner and Schlegelmilch (2017), coded correctly in Blanken et al. (2015). ES was corrected.	ES corrected in the Simbrunner and Schlegelmilch (2017) dataset.
ES coding	Jordan et al. (2011), study 3: ES difference of .03 between the two meta-analyses. Negligible given that $d = 1.00$ in this case.	The ES was recalculated and both ES were coded as $d = .994$.
ES coding	Mazar and Zhong (2010), study 3: recalculated $d = .52$, coded as $d = 3.2$ by Simbrunner and Schlegelmilch (2017). Coded correctly as $d = .53$ in Blanken et al. (2015).	ES corrected in the Simbrunner and Schlegelmilch (2017) dataset to be $d = .53$.

Discrepancy	Study	Action taken
ES coding	Young, Chakroff, and Tom (2012), study 1: $d = -.41$ coded correctly by Simbrunner and Schlegelmilch (2017), coded as $d = .41$ by Blanken et al (2015).	ES corrected in the Blanken et al. (2015) dataset.
N coding	Blanken, van de Ven, and Zeelenberg (2012), study 6: $n = 64$ - first reported in Blanken et al. (2015), coded as $n = 54$ in Simbrunner and Schlegelmilch (2017).	N was replaced in Simbrunner and Schlegelmilch (2017) and the SE was recalculated.
N coding	Blanken et al. (2014), study 4: Simbrunner and Schlegelmilch (2017) coded $n = 614$, Blanken et al. (2015) coded $n = 567$. The original report (Blanken et al. (2014)) does not clearly report $n = 567$.	Despite n being closer to 614 in the original report, we imputed $n = 567$ from the Blanken et al. (2015) meta-analysis.
N coding	Jordan et al. (2011), study 2 & study 3: $n = 68$ and $n = 76$, respectively, coded correctly by Blanken et al. (2015) and coded incorrectly as $n = 84$ and $n = 84$ by Simbrunner and Schlegelmilch (2017).	N was replaced in the Simbrunner and Schlegelmilch (2017) dataset and the SE was recalculated.
N coding	Mazar and Zhong (2010), study 3: $n = 81$ coded as $n = 80$ by Simbrunner and Schlegelmilch (2017).	N was replaced, and SE was recalculated.
N coding	Monin and Miller (2001), study 3: $n = 21$ coded as $n = 20$ by Simbrunner and Schlegelmilch (2017).	N was replaced, and SE was recalculated.
Inclusion	Jordan et al. (2011), study 1: Study 1 was coded by Simbrunner and Schlegelmilch (2017). However, Study 1 only includes moral identity as DV and not behavior. It is hence not a measure of moral licensing.	The study was omitted.
Inclusion	Mazar and Zhong (2010), study 3: Study 3 was coded twice by Simbrunner and Schlegelmilch (2017).	The ES was recalculated (see above) and the second ES was omitted.
Inclusion	Monin and Miller (2001), study 3: Study 3 was coded twice by Blanken et al. (2015).	The second ES was omitted.

Note. Inclusion: The inclusion of the study led to a discrepancy between the meta-analyses.

ES coding & N coding: The effect sizes or the sample sizes were coded inconsistently.

In addition, we examined effects that were included in one meta-analysis but not in the other. In most of these cases, a report was indeed included in only one meta-analysis. In five cases however, the report was included in both meta-analyses, but additional effects were coded in one of them. These additional effect sizes were removed because they either represented duplicates or in one case a study that did not assess moral licensing (see table 1).

Replication of Simbrunner and Schlegelmilch (2017)

In the modified datasets, 100% of the effect sizes and 100% of the sample sizes were identical. Our analyses focus on the larger Simbrunner and Schlegelmilch (2017) dataset, which includes 102 independent effect sizes. First, we attempted to replicate the essential findings reported in the meta-analysis. Analyses were implemented using the metafor-package (Viechtbauer, 2010) of the open source statistics software R (R Core Team, 2016). A random-effects meta-analysis indicated a mean effect size of $d = .25$, $Z = 6.72$, $p < .001$, which is slightly smaller than the $d = .32$ reported by Simbrunner and Schlegelmilch (2017). There was evidence for substantial heterogeneity, $I^2 = 62.26\%$, $Q(101) = 257.63$, $p < .001$. We further replicated the moderator-analysis using culture (Europe vs. North-America vs. South-East Asia) and type of comparison (neutral vs. immoral control condition) as predictors. As reported by Simbrunner and Schlegelmilch (2017), culture had a significant effect, with North-American and European samples showing significantly larger effect sizes than South-East Asian samples, $\beta = .73$, $Z = 5.27$, $p < .001$ and $\beta = .57$, $Z = 4.09$, $p < .001$, respectively. However, the effect of type of comparison - neutral control conditions leading to smaller licensing effects than immoral control conditions - was no longer significant, $\beta = -.14$, $Z = -1.69$, $p = .091$. Therefore, the corrections reported in table 1 reduced the moderating influence of type of comparison, which was substantially higher in the uncorrected dataset, $\beta = -.42$, $Z = -3.72$, $p < .001$. This is largely attributable to the accidental inclusion of two effect sizes with $d > 3$ in the immoral condition.

Publication bias analyses

Over all 102 effect sizes, PET-PEESE indicated a significant positive slope for the standard error, $\beta = 1.47$, $t(100) = 3.07$, $p = .003$. The intercept, representing the corrected effect size, was not different from zero, $\beta = -.10$, $t(100) = -0.93$, $p = .35$ (see figure 1). A 3-PSM over all studies indicated a significant corrected effect size of $d = .10$, $Z = 2.19$, $p = .029$. There was a significant improvement in fit when publication bias was modelled vs. when it was not, $\chi^2(1) = 11.67$, $p < .001$. In order to test the robustness of these findings, we repeated the analyses after simultaneously including relevant moderators. A meta-regression

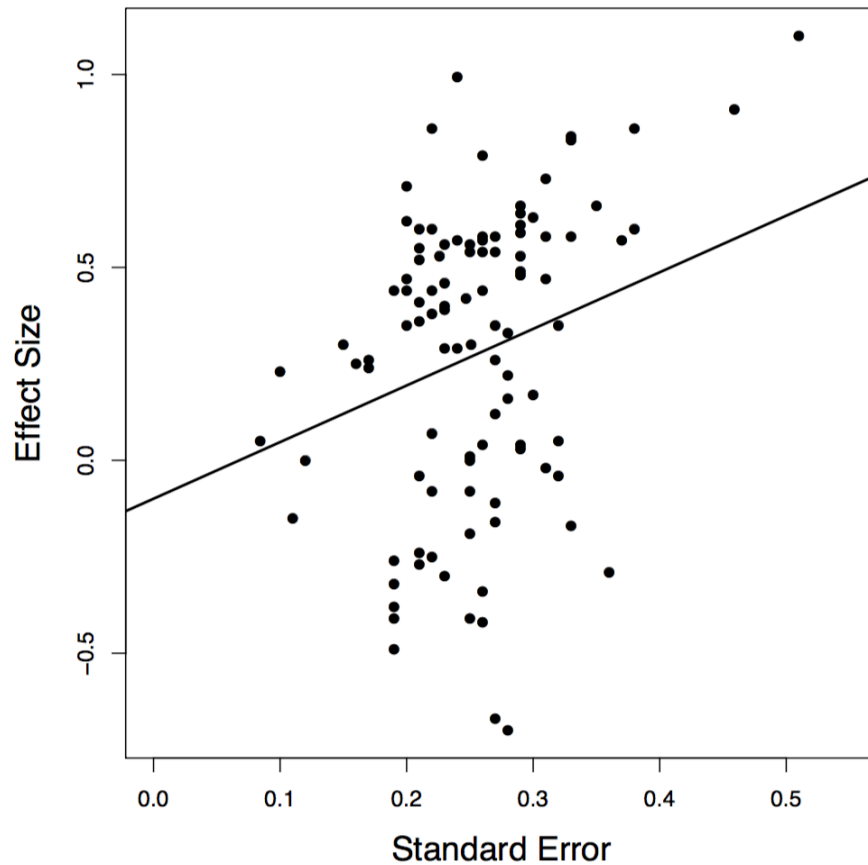


Figure 1. Relationship between standard error and effect size (Cohen's d) of moral licensing for the modified Simbrunner and Schlegelmilch (2017) dataset.

using type of comparison, culture and - resembling PET-PEESE - the standard error as predictors indicated significant effects of world region (North-America vs. South-East-Asia: $\beta = .68$, $Z = 5.01$, $p < .001$; Europe vs. South-East-Asia: $\beta = .51$, $Z = 3.77$, $p < .001$) and the standard error, $\beta = 1.26$, $Z = 2.42$, $p = .015$. Similarly, a 3-PSM including type of comparison and culture as moderators indicated significant effects of culture (North-America vs. South-East Asia: $\beta = .59$, $Z = 4.89$, $p < .001$; Europe vs. South-East-Asia: $\beta = .46$, $Z = 3.92$, $p < .001$), while still showing significant improvement in fit after publication bias is modelled, $\chi^2(1) = 11.55$, $p < .001$. We further repeated the analyses in subgroups formed by dividing groups according to the only significant moderator - culture. Evidence for publication bias was found in both the North-American and the European dataset (see table 2). Both methods to adjust for publication bias resulted in substantial reductions of the mean effect size in the North-American and the European dataset. Given the small number of studies in South-East Asia and the lack of variability in sample size, neither PET-PEESE nor the 3-PSM could be implemented in that dataset.

Table 2

Publication bias analyses.

Sample	Naive rma	PET-PEESE	3-PSM
Entire dataset ($k = 102$)	$d = .25, Z = 6.72, p < .001$	$d = -.10, t(100) = -0.93, p = .35$ $b = 1.47, t(100) = 3.07, p = .003$	$d = .10, Z = 2.19, p = .029$ fit improvement: $\chi^2(1) = 11.67, p < .001$
North America ($k = 48$)	$d = .39, Z = 8.28, p < .001$	$d = -.10, t(46) = -0.87, p = .39$ $b = 2.04, t(46) = 4.00, p < .001$	$d = .27, Z = 3.87, p < .001$ fit improvement: $\chi^2(1) = 4.63, p = .03$
Europe ($k = 44$)	$d = .22, Z = 4.13, p < .001$	$d = .06, t(42) = 0.33, p = .74$ $b = .69, t(42) = 0.93, p = .36$	$d = .08, Z = 1.41, p = .16$ fit improvement: $\chi^2(1) = 6.24, p = .012$
South-East Asia ($k = 5$)	$d = -.37, Z = -4.38, p < .001$	NA	NA

Note. Naive rma = conventional random effects meta-analysis. PET-PEESE: d denotes the intercept / corrected effect size, b denotes the slope / the test for potential publication bias. 3-PSM: d denotes the corrected effect size. The fit-improvement compares the model fit with a baseline model that does not estimate publication bias. NA: Not applicable.

Discussion

Overall, our results indicate strong evidence for publication bias in the moral licensing literature, despite the inclusion of some unpublished studies in the current dataset. This evidence for publication bias persisted after relevant moderators were accounted for using meta-regression and subgroup-analyses. Over all studies, the adjusted effect sizes were $-.10$ (PET-PEESE) and $.10$ (3-PSM), remaining significant in only one of the two analyses. In the subgroup analyses looking at studies from North-America and Europe with two methods to correct for publication bias, the adjusted effect sizes were also substantially reduced. Only one of four effect size estimates remained significantly different from zero.

What can be concluded from this regarding the evidential status of moral licensing? The analyses do not provide unequivocal evidence for an effect size that is significantly different from zero. More importantly, they do not provide evidence for an effect size of practical significance. An effect size of $d = .10$ for the overall sample, which is similar to the $d = .15$ reported by Ebersole et al. (2016), would require $n = 2476$ for 80% power in a one-tailed t-test. It would also imply that studies with an overall $n < 200$ (> 80% of the included studies) would have a power of 17% or less. Using the most optimistic estimate of $d = .27$, which was present only in the 3-PSM model in the North-American subsample, one would require a more achievable $n = 342$ for 80% power, which is larger than the n of 96% of the studies in this subsample.

What remained significant throughout all analyses was the moderating influence of culture. Interestingly, the effect appeared to be entirely reversed in the South-East Asian sample, as reported for the first time by Simbrunner and Schlegelmilch (2017). The significant moderation effect persisted even after the correction for publication bias. Nevertheless, the findings have to be interpreted with caution, given that only 5 experiments were conducted in South-East Asia. All of these experiments are unpublished data from Simbrunner and Schlegelmilch (2017).

Recommendations for further research

If researchers seek to firmly establish the existence of the classical moral licensing effect, more large studies such as the Ebersole (2016) multi-lab project should be pursued. In order to prevent QRPs and publication bias, they should be pre-registered. Using the corrected effect size estimates from this re-analysis, the expected effect would be largest in North-America.

Regarding the moderation by culture, future studies are required to demonstrate this effect. Ideally, these studies should obtain sufficiently large samples in different cultures (e.g., South-East Asia and North-America) and pre-register planned analyses. Based on the very limited present data, the moderation by culture appears to be a more consistent effect than the conventional main effect of moral licensing.

Conclusion

Overall, we found strong evidence for publication bias in the moral licensing literature. The mean effect sizes were often indistinguishable from zero after the correction for publication bias. Large samples would be required to allow for a meaningful study of these effects - sample sizes that were not achieved in the original studies on moral licensing. Future research should aim for high power and employ pre-registration.

References

References marked with an asterisk are exclusively cited in table 1.

- Baumeister, R. F., Bratslavsky, E., Muraven, M., & Tice, D. M. (1998). Ego depletion: Is the active self a limited resource?. *Journal of personality and social psychology*, 74, 1252-1265.
- Becker, B. J. (2005). Failsafe N or file-drawer number. *Publication bias in meta-analysis: Prevention, assessment and adjustments*, 111-125.
- Bem, D. J. (1972). Self-perception theory. *Advances in experimental social psychology*, 6, 1-62.
- *Blanken, I., Van de Ven, N., & Zeelenberg, M. (2012). [A variety of studies on the self-licensing effect]. Unpublished raw data.
- Blanken, I., van de Ven, N., Zeelenberg, M., & Meijers, M. H. (2014). Three attempts to replicate the moral licensing effect. *Social Psychology*, 45, 232-238.
- Blanken, I., van de Ven, N., & Zeelenberg, M. (2015). A meta-analytic review of moral licensing. *Personality and Social Psychology Bulletin*, 41, 540-558.
- Carter, E. C., & McCullough, M. E. (2014). Publication bias and the limited strength model of self-control: has the evidence for ego depletion been overestimated?. *Frontiers in psychology*, 5, 823.
- Carter, E. C., Schönbrodt, F. D., Gervais, W. M., & Hilgard, J. (2018, February 1). Correcting for bias in psychology: A comparison of meta-analytic methods. Retrieved from: <http://doi.org/10.17605/OSF.IO/9H3NU>
- Duval, S., & Tweedie, R. (2000). Trim and fill: a simple funnel- plot-based method of testing and adjusting for publication bias in meta- analysis. *Biometrics*, 56, 455-463.
- Ebersole, C. R., Atherton, O. E., Belanger, A. L., Skulborstad, H. M., Allen, J. M., Banks, J. B., ... & Brown, E. R. (2016). Many Labs 3: Evaluating participant pool quality across the academic semester via replication. *Journal of Experimental Social Psychology*, 67, 68-82.
- *Effron, D. A., Monin, B., & Miller, D. T. (2012). Inventing racist roads not taken: The licensing effect of immoral counterfactual behaviors. *Journal of Personality and Social Psychology*, 103, 916-932.
- Egger, M., Smith, G. D., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *Bmj*, 315, 629-634.
- Etherton, J. L., Osborne, R., Stephenson, K., Grace, M., Jones, C., & De Nadai, A. (2018). Bayesian analysis of multimethod ego- depletion studies favours the null hypothesis. *British Journal of Social Psychology*, 57, 367-385.
- Ferguson, C. J., & Heene, M. (2012). A vast graveyard of undead theories: Publication bias and psychological science's aversion to the null. *Perspectives on Psychological Science*, 7, 555-561.

- Francis, G. (2012). Publication bias and the failure of replication in experimental psychology. *Psychonomic Bulletin & Review*, 19, 975-991.
- Franco, A., Malhotra, N., & Simonovits, G. (2014). Publication bias in the social sciences: Unlocking the file drawer. *Science*, 345, 1502-1505.
- Greenwald, A. G. (1975). Consequences of prejudice against the null hypothesis. *Psychological bulletin*, 82, 1.
- Hagger, M. S., Chatzisarantis, N. L., Alberts, H., Anggono, C. O., Batailler, C., Birt, A. R., ... & Calvillo, D. P. (2016). A multilab preregistered replication of the ego-depletion effect. *Perspectives on Psychological Science*, 11, 546-573.
- Hagger, M. S., Wood, C., Stiff, C., & Chatzisarantis, N. L. (2010). Ego depletion and the strength model of self-control: a meta-analysis. *Psychological bulletin*, 136, 495-525.
- Hedges, L. V. (1984). Estimation of effect size under nonrandom sampling: The effects of censoring studies yielding statistically insignificant mean differences. *Journal of Educational Statistics*, 9, 61-85.
- Hedges, L. V., & Vevea, J. L. (1996). Estimating effect size under publication bias: small sample properties and robustness of a random effects selection model. *Journal of Educational and Behavioral Statistics*, 21, 299-332.
- Ioannidis, J. P., Munafo, M. R., Fusar-Poli, P., Nosek, B. A., & David, S. P. (2014). Publication and other reporting biases in cognitive sciences: detection, prevalence, and prevention. *Trends in cognitive sciences*, 18, 235-241.
- Iyengar, S., & Greenhouse, J. B. (1988). Selection models and the file drawer problem. *Statistical Science*, 109-117.
- *Jordan, J., Mullen, E., & Murnighan, J. K. (2011). Striving for the moral self: The effects of recalling past moral actions on future moral behavior. *Personality and Social Psychology Bulletin*, 37, 701-713.
- Mazar, N., & Zhong, C. B. (2010). Do green products make us better people?. *Psychological science*, 21, 494-498.
- McShane, B. B., Böckenholt, U., & Hansen, K. T. (2016). Adjusting for publication bias in meta-analysis: An evaluation of selection methods and some cautionary notes. *Perspectives on Psychological Science*, 11, 730-749.
- Merritt, A. C., Effron, D. A., & Monin, B. (2010). Moral self- licensing: When being good frees us to be bad. *Social and personality psychology compass*, 4, 344-357.
- Monin, B., & Miller, D. T. (2001). Moral credentials and the expression of prejudice. *Journal of personality and social psychology*, 81, 33-43.
- Moreno, S. G., Sutton, A. J., Ades, A. E., Stanley, T. D., Abrams, K. R., Peters, J. L., & Cooper, N. J. (2009). Assessment of regression-based methods to adjust for publication bias through a comprehensive simulation study. *BMC medical research methodology*, 9, 2.

- Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349, aac4716.
- Pashler, H., & Wagenmakers, E. J. (2012). Editors' introduction to the special section on replicability in psychological science: A crisis of confidence?. *Perspectives on Psychological Science*, 7, 528-530.
- R Core Team (2016). R: A language and environment for statistical computing [Computer software]. Vienna, Austria: R Foundation for Statistical Computing
- Rosenthal, R. (1979). The file drawer problem and tolerance for null results. *Psychological bulletin*, 86, 638-641.
- Rothstein, H. R., Sutton, A. J., & Borenstein, M. (Eds.). (2006). Publication bias in meta-analysis: Prevention, assessment and adjustments. Hoboken, US: John Wiley & Sons.
- Sachdeva, S., Iliev, R., & Medin, D. L. (2009). Sinning saints and saintly sinners: The paradox of moral self-regulation. *Psychological science*, 20, 523-528.
- Simbrunner, P., & Schlegelmilch, B. B. (2017). Moral licensing: a culture-moderated meta-analysis. *Management Review Quarterly*, 67, 201-225.
- Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological science*, 22, 1359-1366.
- Simonsohn, U., Nelson, L. D., & Simmons, J. P. (2014). P-curve: a key to the file-drawer. *Journal of Experimental Psychology: General*, 143, 534-547.
- Stanley, T. D. (2017). Limitations of PET-PEESE and other meta-analysis methods. *Social Psychological and Personality Science*, 8, 581-591.
- Stanley, T. D., & Doucouliagos, H. (2014). Meta- regression approximations to reduce publication selection bias. *Research Synthesis Methods*, 5, 60-78
- Urban, J., Bahník, Š., & Kohlová, M. B. (2017, December 19). Green consumption does not make people cheat: Three replications of a moral licensing experiment. <http://doi.org/10.17605/OSF.IO/WYNJB>
- Vevea, J. L., & Hedges, L. V. (1995). A general linear model for estimating effect size in the presence of publication bias. *Psychometrika*, 60, 419-435.
- Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. *Journal of Statistical Software*, 36, 1-48.
- *Young, Y., Chakroff, A., & Tom, J. (2012). Doing good leads to more good: The reinforcing power of a moral self-concept. *Review of Philosophy and Psychology*, 3, 325-334.